Spectral Lower Bound and an Improved Upper Bound
for the Chromatic Number of Unit Sphere Packings

Abstract

A *unit sphere packing* in $\mathbb{R}^n$ consists of unit spheres of radius $\frac{1}{2}$ with pairwise disjoint interiors, equivalently a set of centres with pairwise distance ≥ 1. The *tangency graph* connects pairs of centres at distance exactly 1, and the *chromatic number* of a packing is the chromatic number of its tangency graph. Let $M(n)$ denote the supremum of chromatic numbers of unit sphere packings in $\mathbb{R}^n$.

Lower bound. We give a self-contained spectral proof that

$$\limsup_{n\to\infty} \frac{M(n)-n}{n^{2/3}} \;\geq\; 1,$$

and more precisely $M(d_q) \geq q^3+1 = d_q + (q^2-q+1)$ for every prime power $q \geq 2$, where $d_q = q^3 - q^2 + q$. The proof combines: (i) a *Spectral Realization Theorem* showing that any connected k-regular graph with least adjacency eigenvalue $s < -1$ of multiplicity g embeds as the tangency graph of a unit sphere packing in $\mathbb{R}^{v-g-1}$, and (ii) the classical *Hoffman ratio bound* $\chi(G) \geq 1 - k/s$. The witness is the complement of the collinearity graph of $\mathrm{GQ}(q, q^2)$.

Upper bound. We prove an improved upper bound

$$M(n) \;\leq\; O\!\left(\tau(n)\cdot\sqrt{\frac{\log n}{n}}\right) \;=\; 2^{0.401n(1+o(1))}\cdot O\!\left(\sqrt{\frac{\log n}{n}}\right),$$

where $\tau(n)$ is the n-dimensional kissing number. This improves the standard greedy bound $M(n) \leq \tau(n)+1$ by a polynomial factor $O(\sqrt{n/\log n}) \to \infty$. The key inputs are: the geometric fact that the clique number of any packing's tangency graph satisfies $\omega \leq n+1$ (equidistant-set bound), the degree bound $\Delta \leq \tau(n)$, and the Bonamy–Kelly–Nelson–Postle theorem on coloring graphs with bounded maximum degree and small clique number.

Barrier. A systematic barrier theorem (Section 6) proves that the four classical spectral families from finite geometry (Kneser graphs, triangular graphs, symplectic polar graphs, and complements of hexagon

point graphs) all have Hoffman bound $H(G) \leq d$, so none can improve the lower-bound surplus beyond $O(d^{2/3})$ within this framework.

Problem

Throughout this paper, a *unit sphere* has radius $\frac{1}{2}$ (diameter 1). Two such spheres have disjoint interiors if and only if their centres are at distance ≥ 1; they are *tangent* when their centres are at distance exactly 1.

A *unit sphere packing* in $\mathbb{R}^n$ is a set of unit spheres (radius $\frac{1}{2}$) with pairwise disjoint interiors, equivalently a set of points (sphere centres) in $\mathbb{R}^n$ with pairwise distance ≥ 1. The *tangency graph* of the packing takes the centres as vertices and connects pairs at distance exactly 1 by an edge. A *colouring* assigns a colour to each sphere so that tangent pairs receive different colours; the *chromatic number* of the packing is the minimum number of colours needed.

Denote by $M(n)$ the supremum of chromatic numbers of all unit sphere packings (finite or infinite) in $\mathbb{R}^n$. The current state of knowledge is:

- *Upper bound*: $M(n) \leq \tau(n) + 1 \leq 2^{(0.401+o(1))n}$ by greedy colouring and the Kabatiansky–Levenshtein bound on the kissing number [4].
- *Lower bound*: $M(n) \geq n + \Omega(n^{2/3})$ for an infinite sequence of dimensions n, first proved by Hao Chen [1] via generalized quadrangles.

We provide: (a) a clean spectral proof of the lower bound establishing the explicit constant $\limsup_{n\to\infty}(M(n)-n)/n^{2/3} \geq 1$; (b) an improved upper bound $M(n) = O(\tau(n)\sqrt{\log n/n})$, a polynomial-factor improvement over greedy; and (c) a barrier theorem for the lower-bound framework.

Main Results

Section 1: Spectral Realization as a Unit Packing

Theorem 0.1 (Spectral Realization)**.** *Let $G = (V, E)$ be a finite connected non-complete k-regular graph on v vertices with adjacency matrix A, least eigenvalue $s < -1$ of multiplicity g, and $d = v - g - 1$. Then there exist points $\{p_x : x \in V\} \subset \mathbb{R}^d$ such that*

$$\|p_x - p_y\| = 1 \quad \text{if } xy \in E, \qquad \|p_x - p_y\| > 1 \quad \text{if } xy \notin E,\ x \neq y.$$

In particular, the tangency graph of the unit sphere packing $\{p_x\}$ (with minimum inter-centre distance 1) is exactly G.

We need to include one extra positive scale factor. The unscaled matrix PMP gives the right **relative** distances, but not generally unit edge length unless $s = -2$. We define the scaled Gram matrix accordingly.

Proof. Let $\mathbf{1} \in \mathbb{R}^v$ be the all-ones vector, let $J = \mathbf{1}\mathbf{1}^\mathsf{T}$, and let

$$P = I - \frac{1}{v}J.$$

Thus P is the orthogonal projection onto $\mathbf{1}^\perp$. Set

$$M = \frac{1}{-s}(A - sI), \qquad \alpha = \frac{s}{2(s+1)}.$$

Since $s < -1$, we have $\alpha > 0$. We will use

$$B = \alpha PMP$$

as the Gram matrix.

Because s is the least eigenvalue of the symmetric matrix A, every eigenvalue of $A - sI$ is nonnegative. Since $-s > 0$, it follows that $M \succeq 0$. Hence for every $z \in \mathbb{R}^v$,

$$z^\mathsf{T} PMPz = (Pz)^\mathsf{T} M(Pz) \geq 0,$$

so $PMP \succeq 0$, and therefore $B \succeq 0$.

Rank computation. Since G is k-regular, $A\mathbf{1} = k\mathbf{1}$. Moreover, because G is connected, the k-eigenspace is exactly $\operatorname{span}\{\mathbf{1}\}$. (Proof: if $Au = ku$ and $u_x := u(x)$ achieves its maximum at some vertex x_0, then $ku_{x_0} = \sum_{y\sim x_0} u_y \leq ku_{x_0}$ with equality only if all neighbours of x_0 also achieve the maximum; by connectedness all entries of u equal u_{x_0}, so $u \in \operatorname{span}\{\mathbf{1}\}$.)

Take an orthogonal eigenbasis of A. The vector $\mathbf{1}$ contributes one eigenvector with eigenvalue k. The eigenspace for the least eigenvalue s has dimension g and lies inside $\mathbf{1}^\perp$ (because for any eigenvector u with $Au = su$ and $s \neq k$, we have $k(1^T u) = (k\mathbf{1})^T u = (A\mathbf{1})^T u = \mathbf{1}^T Au = s(\mathbf{1}^T u)$, so $(k - s)(\mathbf{1}^T u) = 0$, hence $\mathbf{1}^T u = 0$). On $\mathbf{1}^\perp$, the projection P acts as the identity, so if $u \perp \mathbf{1}$ and $Au = \lambda u$, then

$$PMPu = Mu = \frac{\lambda - s}{-s}u.$$

This eigenvalue is 0 exactly when $\lambda = s$, and is positive when $\lambda > s$. Also, $PMP\mathbf{1} = 0$. Therefore,

$$\ker(PMP) = \operatorname{span}\{\mathbf{1}\} \oplus \ker(A - sI),$$

so $\dim\ker(PMP) = 1 + g$, and

$$\operatorname{rank}(B) = \operatorname{rank}(PMP) = v - (1 + g) = v - g - 1 = d.$$

Gram factorization. Since B is a real positive-semidefinite matrix of rank d, it admits a Gram factorisation: let $B = U\Delta U^\mathsf{T}$ be its eigendecomposition, where $\Delta = \operatorname{diag}(\beta_1, \ldots, \beta_d, 0, \ldots, 0)$ with $\beta_1, \ldots, \beta_d > 0$ the positive eigenvalues, and U orthogonal. Set $F = [\sqrt{\beta_1}\, u_1 \mid \cdots \mid \sqrt{\beta_d}\, u_d] \in \mathbb{R}^{v\times d}$, so that $FF^\mathsf{T} = \sum_{i=1}^d \beta_i u_i u_i^\mathsf{T} = B$. For each vertex $x \in V$, define $p_x \in \mathbb{R}^d$ to be the x-th row of F. Then $B_{xy} = \langle p_x, p_y\rangle$.

Distance computation. Let

$$\rho = \frac{k-s}{-s}.$$

Every row sum of M equals ρ: since $\sum_y A_{xy} = k$ (regularity) and $\sum_y I_{xy} = 1$,

$$\sum_y M_{xy} = \frac{1}{-s}(k-s) = \rho.$$

Since M is symmetric, every column sum also equals ρ. Therefore $M\mathbf{1} = \rho\mathbf{1}$, giving $MJ = \rho J$ and $JM = \rho J$, so $JMJ = J(\rho\mathbf{1}\mathbf{1}^T) = \rho v J$. Expanding:

$$PMP = \Big(I - \tfrac{1}{v}J\Big)M\Big(I - \tfrac{1}{v}J\Big) = M - \tfrac{1}{v}MJ - \tfrac{1}{v}JM + \tfrac{1}{v^2}JMJ = M - \tfrac{\rho}{v}J - \tfrac{\rho}{v}J + \tfrac{\rho}{v}J = M - \frac{\rho}{v}J.$$

For every vertex x, $M_{xx} = \frac{-s}{-s} = 1$. For distinct vertices $x \neq y$,

$$M_{xy} = \begin{cases} \dfrac{1}{-s}, & xy \in E, \\ 0, & xy \notin E. \end{cases}$$

Therefore $(B_0)_{xx} = 1 - \frac{\rho}{v}$ (where $B_0 = PMP$), and for $x \neq y$,

$$(B_0)_{xy} = \begin{cases} -\dfrac{1}{s} - \dfrac{\rho}{v}, & xy \in E, \\ -\dfrac{\rho}{v}, & xy \notin E. \end{cases}$$

Now let $x \neq y$. Since $B = \alpha B_0$ is the Gram matrix of the p_x's,

$$\|p_x - p_y\|^2 = B_{xx} + B_{yy} - 2B_{xy}.$$

If $xy \in E$:

$$\|p_x - p_y\|^2 = \alpha\Big[2\Big(1 - \tfrac{\rho}{v}\Big) - 2\Big(-\tfrac{1}{s} - \tfrac{\rho}{v}\Big)\Big] = \alpha\Big(2 + \tfrac{2}{s}\Big) = \frac{s}{2(s+1)} \cdot \frac{2(s+1)}{s} = 1.$$

Thus $\|p_x - p_y\| = 1$ whenever $xy \in E$.

If $xy \notin E$, $x \neq y$:

$$\|p_x - p_y\|^2 = \alpha\Big[2\Big(1 - \tfrac{\rho}{v}\Big) - 2\Big(-\tfrac{\rho}{v}\Big)\Big] = 2\alpha = \frac{s}{s+1}.$$

Since $s < -1$, write $s = -1 - \varepsilon$ with $\varepsilon > 0$; then $\frac{s}{s+1} = \frac{1+\varepsilon}{\varepsilon} > 1$. Hence $\|p_x - p_y\| > 1$ whenever $xy \notin E$ and $x \neq y$.

Therefore the point set $\{p_x : x \in V\} \subset \mathbb{R}^d$ has minimum pairwise distance at least 1, with equality exactly on edges of G. Consequently, the tangency graph of this unit sphere packing is exactly G. $\square$

Section 2: Hoffman Lower Bound for Chromatic Number

Theorem 0.2 (Hoffman Bound)**.** *Let G be a finite k-regular graph on v vertices with least adjacency eigenvalue $s < 0$. Then*

$$\chi(G) \;\geq\; 1 - \frac{k}{s}.$$

Proof. For reference, the inequality to be proved is the standard Hoffman lower bound on the chromatic number: $\chi(G) \geq 1 - \lambda_1/\lambda_n$, where λ_1 and λ_n are the largest and smallest adjacency eigenvalues. We prove the stated k-regular case directly.

Let A be the adjacency matrix of G, and let $\mathbf{1} \in \mathbb{R}^v$ be the all-ones vector. Since G is k-regular, $A\mathbf{1} = k\mathbf{1}$.

Let $S \subseteq V(G)$ be any independent set, and let $\mathbf{x}$ be its indicator vector. Put $n := |S| = \mathbf{x}^T\mathbf{x}$. If $n = 0$, there is nothing to prove, so assume $n > 0$.

Decompose $\mathbf{x}$ into its component along $\mathbf{1}$ and its orthogonal component:

$$\mathbf{x} = \frac{n}{v}\mathbf{1} + \mathbf{u}, \qquad \mathbf{u} \perp \mathbf{1}.$$

Indeed, $\mathbf{1}^T\mathbf{u} = \mathbf{1}^T\mathbf{x} - \frac{n}{v}\mathbf{1}^T\mathbf{1} = n - \frac{n}{v}v = 0$.

Because A is real symmetric, it has an orthonormal eigenbasis. Every eigenvalue of A is at least s, so for every vector $\mathbf{u} \perp \mathbf{1}$ we have $\mathbf{u}^T A\mathbf{u} \geq s\|\mathbf{u}\|^2$.

Now compute $\mathbf{x}^T A\mathbf{x}$. Since S is independent, no edge has both endpoints in S, so $\mathbf{x}^T A\mathbf{x} = 0$. On the other hand,

$$\mathbf{x}^T A\mathbf{x} = \left(\frac{n}{v}\mathbf{1} + \mathbf{u}\right)^T A\left(\frac{n}{v}\mathbf{1} + \mathbf{u}\right) = \frac{n^2}{v^2}\mathbf{1}^T A\mathbf{1} + \frac{2n}{v}\mathbf{1}^T A\mathbf{u} + \mathbf{u}^T A\mathbf{u}.$$

Since $A\mathbf{1} = k\mathbf{1}$ and $\mathbf{u} \perp \mathbf{1}$, we have $\mathbf{1}^T A\mathbf{1} = kv$ and $\mathbf{1}^T A\mathbf{u} = (A\mathbf{1})^T\mathbf{u} = k\mathbf{1}^T\mathbf{u} = 0$. Therefore

$$0 = \mathbf{x}^T A\mathbf{x} = \frac{kn^2}{v} + \mathbf{u}^T A\mathbf{u}.$$

Using $\mathbf{u}^T A\mathbf{u} \geq s\|\mathbf{u}\|^2$ gives

$$0 \geq \frac{kn^2}{v} + s\|\mathbf{u}\|^2.$$

Now $\|\mathbf{u}\|^2 = \|\mathbf{x} - \frac{n}{v}\mathbf{1}\|^2 = \|\mathbf{x}\|^2 - \frac{n^2}{v} = n - \frac{n^2}{v} = \frac{n(v-n)}{v}$. Hence

$$0 \geq \frac{kn^2}{v} + s\frac{n(v-n)}{v}.$$

Multiplying by $\frac{v}{n} > 0$ yields

$$0 \geq kn + s(v-n) = (k-s)n + sv.$$

Thus $(k-s)n \leq -sv$. Since $s < 0$ and $k - s > 0$, we obtain

$$n \leq \frac{-sv}{k-s}.$$

So every independent set in G has size at most $\frac{-sv}{k-s}$.

Now let $V_1, \ldots, V_\chi$ be the color classes of an optimal proper coloring of G, where $\chi = \chi(G)$. Each V_i is an independent set, so $|V_i| \le \frac{-sv}{k-s}$ for each i. Since the color classes partition $V(G)$,

$$v = \sum_{i=1}^{\chi} |V_i| \le \chi \cdot \frac{-sv}{k-s}.$$

Dividing by $v > 0$ gives $1 \le \chi \cdot \frac{-s}{k-s}$. Therefore

$$\chi \ge \frac{k-s}{-s} = 1 - \frac{k}{s}. \qquad \square$$

$\square$

Section 3: The Generalized Quadrangle Construction

Theorem 0.3 (Spectral Family from $\mathrm{GQ}(q, q^2)$)**.** *For every prime power $q \ge 2$, let G_q denote the complement of the point-collinearity graph of the generalized quadrangle* $\mathrm{GQ}(q, q^2)$*. Then:*

1. *G_q has $v_q = (q+1)(q^3+1)$ vertices and is regular of degree $k_q = q^4$.*
2. *The least adjacency eigenvalue of G_q is $s_q = -q$, with multiplicity $g_q = q^4 + q^2$, and the embedding dimension is $d_q := v_q - g_q - 1 = q^3 - q^2 + q$.*
3. *The Hoffman lower bound satisfies*
$$1 - \frac{k_q}{s_q} = q^3 + 1 = d_q + (q^2 - q + 1).$$
4. *For every prime power $q \ge 2$,*
$$1 - \frac{k_q}{s_q} \ge d_q + \left(\frac{3}{4}\right)^{1/3} d_q^{2/3}.$$

Proof. Let $\mathcal{Q}$ be a generalized quadrangle of order (q, q^2), and let Γ_q be its point-collinearity graph. Thus two distinct points are adjacent in Γ_q exactly when they are collinear in $\mathcal{Q}$. Generalized quadrangles of order (q, q^2) exist for all prime powers q: the classical realization is the *elliptic quadric* $Q^-(5, q)$ in $\mathrm{PG}(5, q)$, which has $(q^3+1)(q+1)$ points and forms a $\mathrm{GQ}(q, q^2)$ (see [5] and [2]). The Hermitian variety $H(3, q^2)$ gives $\mathrm{GQ}(q^2, q)$, not $\mathrm{GQ}(q, q^2)$.

The collinearity graph of a $\mathrm{GQ}(s, t)$ is strongly regular with parameters [5]

$$v = (s+1)(st+1), \quad k = s(t+1), \quad \lambda = s - 1, \quad \mu = t + 1,$$

and nontrivial eigenvalues $s-1$ (multiplicity $st(s+1)(t+1)/(s+t)$) and $-(t+1)$ (multiplicity $s^2(st+1)/(s+t)$).

Setting $s = q$ and $t = q^2$:

$$v = (q+1)(q^3+1), \quad \kappa = q(q^2+1), \quad \lambda = q-1, \quad \mu = q^2+1,$$

with nontrivial eigenvalues

$$r = q-1, \qquad s_0 = -(q^2+1).$$

Connectedness. If two points are collinear, they are adjacent in Γ_q. If two points are not collinear, they have exactly $\mu = q^2+1 > 0$ common neighbours. Hence every pair of vertices is at distance at most 2 in Γ_q, so Γ_q is connected. For any connected k-regular graph, the degree eigenvalue k has multiplicity 1: if $Au = \kappa u$ then the maximum-entry argument (same as in Section 1) shows $u \in \operatorname{span}\{\mathbf{1}\}$, so the κ-eigenspace is 1-dimensional.

Multiplicity computation. Let the multiplicities of r and s_0 be f and g_0 respectively. The trace of the adjacency matrix of a simple graph (no self-loops) equals 0 since $\operatorname{tr}(A) = \sum_x A_{xx} = 0$; by the spectral theorem, this equals the sum of eigenvalues with multiplicity: $\kappa \cdot 1 + r \cdot f + s_0 \cdot g_0 = 0$. Combined with $1 + f + g_0 = v$, the trace condition is $\kappa + (q-1)f - (q^2+1)g_0 = 0$. Using $g_0 = v - 1 - f$:

$$\kappa + (q-1)f - (q^2+1)(v-1-f) = 0 \implies (q^2+q)f = (q^2+1)(v-1) - \kappa.$$

Substituting $v - 1 = q^4 + q^3 + q$ and $\kappa = q(q^2+1)$:

$$f = \frac{(q^2+1)(q^4+q^3+q) - q(q^2+1)}{q^2+q} = \frac{(q^2+1)(q^4+q^3)}{q(q+1)} = q^2(q^2+1) = q^4+q^2.$$

Hence $g_0 = v - 1 - f = (q^4+q^3+q) - (q^4+q^2) = q^3 - q^2 + q$.

Complement eigenvalues. Let $G_q = \overline{\Gamma_q}$ with adjacency matrix $B = J - I - A$. Since Γ_q is κ-regular, $B\mathbf{1} = (v-1-\kappa)\mathbf{1}$, so G_q has degree

$$k_q = v - 1 - \kappa = (q^4+q^3+q) - (q^3+q) = q^4.$$

This proves item (1).

For the non-principal eigenvalues: if $Au = \theta u$ with $\theta \neq \kappa$, then $\kappa(\mathbf{1}^T u) = \mathbf{1}^T A u = \theta(\mathbf{1}^T u)$, so $\mathbf{1}^T u = 0$. Thus all non-principal eigenvectors lie in $\mathbf{1}^\perp$, and for $u \perp \mathbf{1}$ with $Au = \theta u$: $Bu = (J - I - A)u = 0 - u - \theta u = (-1-\theta)u$. Hence G_q has eigenvalues:

- q^4 (multiplicity 1, principal eigenvalue),
- $-r - 1 = -q$ (multiplicity $f = q^4 + q^2$),
- $-s_0 - 1 = q^2$ (multiplicity $g_0 = q^3 - q^2 + q$).

For $q \geq 2$, the three eigenvalues satisfy $q^4 > 0$, $q^2 > 0$, and $-q < 0$. Hence the least adjacency eigenvalue of G_q is $s_q = -q$ with multiplicity $g_q = f = q^4 + q^2$.

Embedding dimension.

$$d_q := v_q - g_q - 1 = (q^4 + q^3 + q + 1) - (q^4 + q^2) - 1 = q^3 - q^2 + q.$$

Non-completeness and connectivity of G_q. We have $k_q = q^4 < q^4 + q^3 + q = v_q - 1$, so G_q is not complete.

For connectivity, recall that two vertices x, y are *adjacent in* G_q if and only if they are *non-collinear in* $\mathcal{Q}$. We show every pair $x \neq y$ has a common neighbour in G_q, establishing diameter at most 2.

Case 1: x and y are adjacent in G_q (non-collinear in $\mathcal{Q}$). Then xy is already an edge, and they trivially have a common neighbour if we can find $z \neq x, y$ adjacent to both; the diameter-2 bound suffices for connectivity, but adjacency already gives a path.

Case 2: x and y are non-adjacent in G_q (collinear in $\mathcal{Q}$). We need a vertex z adjacent to both x and y in G_q, i.e., non-collinear with both. In $\mathcal{Q}$, each point has $\kappa = q(q^2+1)$ collinear points, so x is non-collinear with exactly $v-1-\kappa = q^4$ points of $\mathcal{Q} \setminus \{x\}$; call this set A. Similarly let B be the q^4 points non-collinear with y. Since x and y are collinear, $x \notin A$ and $y \notin B$ (a point is always collinear with itself, but the exclusion $\{x, y\}$ merely removes two points from the ambient set). More precisely, $A \subseteq V \setminus \{x\}$ and $B \subseteq V \setminus \{y\}$, and since x is collinear with y we have $x \notin B$ and $y \notin A$. So both A and B are subsets of $V \setminus \{x, y\}$, which has $v_q - 2 = q^4 + q^3 + q - 1$ elements.

Since $|A| + |B| = 2q^4 > q^4 + q^3 + q - 1 = |V \setminus \{x, y\}|$ for all $q \geq 2$ (because $q^4 - (q^3+q-1) = (q-1)(q^3-1) > 0$ for $q \geq 2$), by inclusion-exclusion $A \cap B \neq \emptyset$. Any $z \in A \cap B$ is non-collinear with both x and y, so z is adjacent to both in G_q, giving the path x-z-y.

Thus G_q has diameter at most 2 and is connected. This proves item (2).

Hoffman bound.

$$1 - \frac{k_q}{s_q} = 1 - \frac{q^4}{-q} = 1 + q^3 = q^3 + 1.$$

Since $d_q + (q^2 - q + 1) = (q^3 - q^2 + q) + (q^2 - q + 1) = q^3 + 1$, we have

$$1 - \frac{k_q}{s_q} = d_q + (q^2 - q + 1).$$

This proves item (3).

Explicit surplus bound. Let $A_q = q^2 - q + 1$, so $d_q = qA_q$ and the surplus equals A_q. It suffices to show $A_q \geq (3/4)^{1/3} d_q^{2/3} = (3/4)^{1/3}(qA_q)^{2/3}$. Since $A_q > 0$, cubing gives the equivalent inequality $A_q^3 \geq (3/4)q^2 A_q^2$, i.e. $A_q \geq (3/4)q^2$. Computing:

$$A_q - \frac{3}{4}q^2 = q^2 - q + 1 - \frac{3}{4}q^2 = \frac{1}{4}q^2 - q + 1 = \frac{(q-2)^2}{4} \geq 0.$$

Therefore $q^2 - q + 1 \geq (3/4)^{1/3} d_q^{2/3}$ for all q, and item (4) follows. $\square$

Section 4: Packing Lower Bounds at the Special Dimensions

Theorem 0.4 (Main Packing Bound)**.** *For every prime power $q \geq 2$, the supremum $M(d_q)$ of chromatic numbers of finite unit sphere packings in $\mathbb{R}^{d_q}$ satisfies*

$$M(d_q) \;\geq\; q^3 + 1 \;=\; d_q + (q^2 - q + 1) \;\geq\; d_q + \left(\frac{3}{4}\right)^{1/3} d_q^{2/3},$$

where $d_q = q^3 - q^2 + q$.

Proof. Fix a prime power $q \geq 2$. Let G_q be the graph from Section 3. We explicitly verify that G_q meets all the hypotheses of the Spectral Realization Theorem (Section 1):

- *Finite*: G_q has $v_q = (q+1)(q^3+1)$ vertices (Section 3, item 1).
- *Connected*: proved in Section 3 (diameter-at-most-2 argument via the $|A \cap B| \neq \emptyset$ counting for non-adjacent pairs).
- *Non-complete*: $k_q = q^4 < v_q - 1$ (Section 3, item 1).
- *k_q-regular*: G_q is regular of degree $k_q = q^4$ (Section 3, item 1).
- *Least eigenvalue $s_q = -q < -1$*: since $q \geq 2$, we have $s_q = -q \leq -2 < -1$ (Section 3, item 2).
- *Multiplicity g_q*: $g_q = q^4 + q^2$ (Section 3, item 2).
- *Dimension formula*: $d_q = v_q - g_q - 1 = q^3 - q^2 + q$ (Section 3, item 2).

By the Spectral Realization Theorem, there exist points $\{p_x : x \in V(G_q)\} \subset \mathbb{R}^{d_q}$ forming a unit sphere packing in $\mathbb{R}^{d_q}$ whose tangency graph is exactly G_q. Therefore

$$M(d_q) \geq \chi(G_q).$$

Since G_q is k_q-regular with least eigenvalue $s_q = -q < 0$, the Hoffman Bound (Section 2) gives

$$\chi(G_q) \geq 1 - \frac{k_q}{s_q} = 1 + q^3 = q^3 + 1.$$

By Section 3, item 3 and item 4:

$$q^3 + 1 = d_q + (q^2 - q + 1) \geq d_q + \left(\frac{3}{4}\right)^{1/3} d_q^{2/3}.$$

Combining: $M(d_q) \geq \chi(G_q) \geq q^3 + 1 \geq d_q + (3/4)^{1/3} d_q^{2/3}$. ☐

Section 5: Monotonicity and the Limsup Lower Bound

Theorem 0.5 (Monotonicity and Limsup Bound)**.** *Let $M(n)$ denote the supremum of chromatic numbers of unit sphere packings (finite or infinite) in $\mathbb{R}^n$. Then:*

1. ***(Monotonicity)*** *$M(n+1) \geq M(n)$ for all $n \geq 1$.*
2. ***(Limsup lower bound)***

$$\limsup_{n\to\infty} \frac{M(n)-n}{n^{2/3}} \geq 1.$$

Remark 0.6. The constant 1 is approached but not necessarily achieved: the proof gives $(M(d_q)-d_q)/d_q^{2/3} \geq (1-1/q+1/q^2)^{1/3}$, which tends to 1 from below as $q\to\infty$. No upper bound of the form $M(n)-n = O(n^{2/3})$ is claimed here.

Remark 0.7. The gap $d_{q+1}-d_q = 3q^2+q+1 \sim 3d_q^{2/3}$ is of the same order as the surplus $q^2-q+1 \sim d_q^{2/3}$, so a naive monotonicity interpolation does *not* extend the bound $M(n) \geq n + c\,n^{2/3}$ to all large n from this sequence alone.

Proof. **(1) Monotonicity.** Define the embedding $\iota : \mathbb{R}^n \to \mathbb{R}^{n+1}$ by $\iota(p) = (p,0)$. For any unit sphere packing $\mathcal{P}$ in $\mathbb{R}^n$ (finite or infinite), the image $\iota(\mathcal{P}) = \{(p,0) : p \in \mathcal{P}\}$ is a unit sphere packing in $\mathbb{R}^{n+1}$, because

$$\|\iota(p)-\iota(q)\|_{\mathbb{R}^{n+1}} = \|p-q\|_{\mathbb{R}^n}$$

for all $p,q \in \mathcal{P}$. In particular, distances are preserved, so tangencies are preserved, and the tangency graph of $\iota(\mathcal{P})$ is isomorphic to that of $\mathcal{P}$. Hence every chromatic number achievable by a packing in $\mathbb{R}^n$ is also achievable by a packing in $\mathbb{R}^{n+1}$, giving $M(n) \leq M(n+1)$.

(2) Limsup bound. By the Main Packing Bound (Section 4), for every prime power $q \geq 2$:

$$M(d_q) \;\geq\; q^3+1 \;=\; d_q + (q^2-q+1), \qquad d_q = q^3-q^2+q.$$

Since $d_q = q(q^2-q+1)$, we have $d_q^{2/3} = q^{2/3}(q^2-q+1)^{2/3}$, so

$$\frac{M(d_q)-d_q}{d_q^{2/3}} \;\geq\; \frac{q^2-q+1}{d_q^{2/3}} = \frac{(q^2-q+1)^{1/3}}{q^{2/3}} = \left(1-\frac{1}{q}+\frac{1}{q^2}\right)^{1/3}.$$

The function $f(q) = 1-1/q+1/q^2$ satisfies $f'(q) = 1/q^2-2/q^3 = (q-2)/q^3 \geq 0$ for $q \geq 2$, so f is nondecreasing for $q \geq 2$ and $\lim_{q\to\infty} f(q) = 1$.

Fix any $\varepsilon \in (0,1)$. Since $f(q) \to 1$, there exists a prime power q_0 such that $f(q_0) > 1-\varepsilon$. By Euclid's theorem on the infinitude of primes, there exist infinitely many prime powers $q > q_0$, giving infinitely many indices $n_j = d_{q_j} \to \infty$ with

$$\frac{M(n_j)-n_j}{n_j^{2/3}} \;\geq\; f(q_j)^{1/3} \;>\; (1-\varepsilon)^{1/3}.$$

Therefore $\limsup_{n\to\infty}(M(n)-n)/n^{2/3} \ge (1-\varepsilon)^{1/3}$ for every $\varepsilon \in (0,1)$. Letting $\varepsilon \to 0$ gives

$$\limsup_{n\to\infty} \frac{M(n)-n}{n^{2/3}} \ \ge\ 1. \qquad \square$$

$\square$

Section 6: Barrier Theorem for the Hoffman Framework

The main result gives $M(n) \ge n+\Omega(n^{2/3})$ via the Spectral Realization Theorem and the Hoffman bound. A natural question is whether the same framework can yield a larger surplus of order n^α for some $\alpha > 2/3$. This section shows that for all standard graph families from finite geometry the answer is no.

We collect the framework parameters. For a connected k-regular graph G with v vertices and least eigenvalue $s < -1$ of multiplicity g:

- embedding dimension: $d = v - g - 1$,
- Hoffman lower bound: $H(G) := 1 - k/s$,
- surplus: $H(G) - d$.

Remark 0.8. Throughout this section, "standard spectral families from finite geometry" refers to exactly the four families listed below: Kneser graphs, triangular graphs, non-collinearity graphs of symplectic polar spaces, and complements of generalized-hexagon point graphs. The theorem makes no claim about other families.

Theorem 0.9 (Hoffman-Framework Barrier)**.** *For the following four families, the Hoffman bound does not exceed d:*

1. (Kneser $K(n,k)$, fixed $k \ge 3$): $H = n/k$, $d \sim n^k/k!$, *so* $H = o(d)$.
2. (Triangular $T(n)$, $n \ge 5$): $H = d = n-1$, *surplus* $= 0$.
3. (Non-collinearity of $W(2r-1,q)$, $r \ge 2$): $H \le d$, *equality only at* $(r,q) = (2,2)$.
4. (Hexagon complement, order (q,q)): $H \sim \frac{1}{2}q^4$, $d \sim \frac{5}{6}q^5$, *so* $H = o(d)$.

None of these families achieves $H(G) \ge d + c\,d^\alpha$ for any $c > 0$, $\alpha > 0$.

Proof. **(1) Kneser graph $K(n,k)$, fixed $k \ge 3$, $n > 2k$.** The Kneser graph $K(n,k)$ has vertex set $\binom{[n]}{k}$ (all k-subsets of $[n]$), adjacent when disjoint. For $n > 2k$, its adjacency eigenvalues are

$$\lambda_j = (-1)^j \binom{n-k-j}{k-j}, \quad 0 \le j \le k,$$

with multiplicities $\binom{n}{j} - \binom{n}{j-1}$ for $j \geq 1$ [2]. In particular the $j = 1$ term,

$$\lambda_1 = -\binom{n-k-1}{k-1},$$

has multiplicity $\binom{n}{1} - \binom{n}{0} = n - 1$.

Claim: λ_1 is the least eigenvalue for every $n > 2k$, $k \geq 3$.

Proof: The even-j eigenvalues are positive for $n > 2k$. For odd $j \geq 3$, compare with $j = 1$:

$$\frac{|\lambda_1|}{|\lambda_j|} = \frac{\binom{n-k-1}{k-1}}{\binom{n-k-j}{k-j}} = \prod_{i=1}^{j-1} \frac{n-k-i}{k-i}.$$

For each $i \in \{1, \ldots, j-1\}$: since $j \leq k$ implies $i \leq k-2$, the denominator $k - i \geq 2 > 0$. Since $n > 2k$, we have $n - k - i > k - i$ (equivalently $n > 2k$), so each factor exceeds 1. Hence the product exceeds 1, giving $|\lambda_j| < |\lambda_1|$ for every odd $j \geq 3$. Thus $s_G = \lambda_1$ with $g = n - 1$. $\square$

The degree is $k_G = \binom{n-k}{k}$, so the Hoffman bound $H = 1 + \binom{n-k}{k} / \binom{n-k-1}{k-1} = n/k$, and $d = \binom{n}{k} - n \sim n^k/k!$. For fixed $k \geq 3$: $H/d \to 0$.

(2) Triangular graph $T(n)$, $n \geq 5$. The triangular graph $T(n)$ has vertex set $\binom{[n]}{2}$ with two vertices adjacent when they share one element; $v = \binom{n}{2}$, $k_G = 2(n-2)$. Two distinct 2-subsets of $[n]$ either are disjoint or share one element, so $T(n) = \overline{K(n,2)}$.

We derive the spectrum of $T(n)$ from the formula of Part (1) applied to $K(n,2)$ (with $k = 2$ [2]):

- $j = 1$: $\lambda_1 = -\binom{n-3}{1} = -(n-3)$, multiplicity $n - 1$.
- $j = 2$: $\lambda_2 = \binom{n-4}{0} = 1$, multiplicity $\binom{n}{2} - n$.

Each nontrivial eigenvector of $K(n,2)$ lies in $\mathbf{1}^\perp$ and becomes an eigenvector of $T(n) = \overline{K(n,2)}$ with eigenvalue $-1 - \lambda$, giving:

$$s_G = -1 - 1 = -2 \;\; (\text{mult } g = \tbinom{n}{2} - n), \qquad n - 4 \;\; (\text{mult } n - 1).$$

Hence $d = \binom{n}{2} - (\binom{n}{2} - n) - 1 = n - 1$ and $H = 1 - 2(n-2)/(-2) = n - 1 = d$; surplus $= 0$.

(3) Non-collinearity graph of $W(2r-1, q)$, $r \geq 2$. The symplectic polar space $W(2r-1, q)$ has $v = (q^{2r} - 1)/(q - 1)$ points. Its non-collinearity graph G is strongly regular with degree $k_G = q^{2r-1}$ and nontrivial eigenvalues $\pm q^{r-1}$ [2]. From the trace equations $f + g = v - 1$ and $f - g = -k_G/q^{r-1} = -q^r$:

$$g = \tfrac{v-1+q^r}{2}, \quad d = v - g - 1 = \tfrac{v-1-q^r}{2}, \quad H = 1 + q^r.$$

We show $d \geq H$, i.e. $v \geq 3 + 3q^r$, by three cases.

Case A: $r = 2$. $v = q^3 + q^2 + q + 1$, so $v - 3 - 3q^2 = q^3 - 2q^2 + q - 2 = (q-2)(q^2+1) \geq 0$, with equality iff $q = 2$. At $(r, q) = (2, 2)$: $d = H = 5$.

Case B: $r = 3$, $q = 2$. Direct computation: $v = (2^6 - 1)/(2 - 1) = 63$, $d = (63 - 1 - 8)/2 = 27$, $H = 1 + 8 = 9$; $d = 27 > H = 9$.

Case C: $r \geq 3$, $(r, q) \neq (3, 2)$. Here $q^{r-1} > 4$: if $r = 3$ then $q \geq 3$ gives $q^2 \geq 9$; if $r \geq 4$ then $q^{r-1} \geq 2^3 = 8$. Using $v \geq q^{2r-1}$ and $3 + 3q^r \leq 4q^r$:

$$v - 3 - 3q^r \geq q^{2r-1} - 4q^r = q^r(q^{r-1} - 4) > 0.$$

Thus $d > H$ in all cases except $(r, q) = (2, 2)$ where $d = H$.

(4) Complement of the generalized hexagon point graph, order (q, q). Let Γ be the point-collinearity graph of a generalized hexagon of order (q, q) [2]: $v = (q+1)(q^4+q^2+1)$, degree $q(q+1)$, nontrivial eigenvalues $2q-1$ (mult m_1), -1 (mult m_2), $-(q+1)$ (mult m_3), where

$$m_1 = \tfrac{q(q+1)^2(q^2+q+1)}{6}, \quad m_2 = \tfrac{q(q+1)^2(q^2-q+1)}{2}, \quad m_3 = \tfrac{q(q^2+q+1)(q^2-q+1)}{3}.$$

For $G = \overline{\Gamma}$: $k_G = q^3(q^2 + q + 1)$, and the nontrivial eigenvalues of G are $-2q$ (mult m_1), 0 (mult m_2), q (mult m_3). The least eigenvalue is $s_G = -2q$ with multiplicity $g = m_1$, so

$$d = v - m_1 - 1 = m_2 + m_3 \sim \tfrac{5}{6}q^5, \qquad H = 1 + \tfrac{q^2(q^2+q+1)}{2} \sim \tfrac{1}{2}q^4.$$

Thus $H/d \sim 3/(5q) \to 0$. Explicit checks: for $q = 2$, $(m_1, m_2, m_3) = (21, 27, 14)$, $v = 63$, $d = 41$, $H = 15$; for $q = 3$, $d = 259$, $H = 59.5$. In all cases $H < d$.

Summary. For Kneser ($k \geq 3$) and the hexagon complement, $H = o(d)$; for the triangular graph and symplectic cases, $H \leq d$ with equality only at $(r, q) = (2, 2)$. None achieves $H \geq d + c\,d^\alpha$ for any $c, \alpha > 0$. □

Remark 0.10. Any improvement to the surplus exponent $\alpha = 2/3$ within the Hoffman framework requires either a genuinely new graph family or a proof technique beyond the Hoffman ratio bound.

Section 7: Improved Upper Bound via Small-Clique Graph Coloring

The following theorem improves the standard upper bound $M(n) \leq \tau(n)+1$ by a polynomial factor, where $\tau(n)$ denotes the n-dimensional kissing number.

Theorem 0.11 (Improved Upper Bound)**.** *There exists an absolute constant $C > 0$ such that, for all sufficiently large n,*

$$M(n) \;\leq\; C\,\tau(n)\sqrt{\frac{\ln(n+2)}{\ln \tau(n)}}.$$

In particular, using the Kabatiansky–Levenshtein bound $\tau(n) \leq 2^{0.401n(1+o(1))}$,

$$M(n) \;=\; O\!\left(\tau(n)\cdot\sqrt{\frac{\log n}{n}}\right) \;=\; 2^{0.401n(1+o(1))}\cdot O\!\left(\sqrt{\frac{\log n}{n}}\right).$$

This is a polynomial-factor improvement over the greedy bound $M(n) \leq \tau(n)+1$.

The proof rests on three ingredients, which we state as lemmas.

Lemma 0.12 (de Bruijn–Erdős Compactness)**.** *$M(n) = \sup\{\chi(G_X) : X \subset \mathbb{R}^n$ finite, $\|x-y\| \geq 1\ \forall x \neq y \in X\}$. In other words, $M(n)$ is achieved as a supremum over* finite *packings.*

Proof. Let $\mathcal{P}$ be any unit sphere packing in $\mathbb{R}^n$ (possibly infinite), and let $G = G_{\mathcal{P}}$ be its tangency graph. Every finite subgraph of G is the tangency graph of a finite sub-packing, hence has chromatic number at most $\sup_{\text{finite } X} \chi(G_X) =: K$. By the de Bruijn–Erdős theorem [6], an infinite graph whose every finite subgraph is K-colorable is itself K-colorable, so $\chi(G) \leq K$. Hence $M(n) \leq K$; the reverse inequality $M(n) \geq K$ is trivial. □

Lemma 0.13 (Geometric Graph Bounds)**.** *For every finite unit sphere packing $X \subset \mathbb{R}^n$ (centres with pairwise distance ≥ 1), the tangency graph G_X satisfies:*

(a) (Degree bound) $\Delta(G_X) \leq \tau(n)$*, the n-dimensional kissing number.*

(b) (Clique bound) $\omega(G_X) \leq n+1$.

Proof. **(a).** The neighbours of a centre $v \in X$ in G_X are the centres $u \in X$ with $\|u-v\| = 1$, all lying on the unit sphere $S^{n-1}(v)$ of radius 1 centred at v, and pairwise at distance ≥ 1. Translating to $v = 0$, this is a set of unit-sphere points with pairwise angular separation ≥ 60ř, i.e. a kissing configuration. By definition of $\tau(n)$, the size of any kissing configuration is at most $\tau(n)$.

(b). A k-clique in G_X is a set $\{x_1, \ldots, x_k\} \subset X$ with $\|x_i - x_j\| = 1$ for all $i \neq j$ — an equidistant set of diameter 1 in $\mathbb{R}^n$. Setting $y_i = x_i - x_1$ for $i \geq 2$, we have $\|y_i\| = 1$ and

$$\|y_i - y_j\|^2 = \|x_i - x_j\|^2 = 1, \quad i \neq j \geq 2,$$

which gives $\langle y_i, y_j \rangle = \frac{1}{2}$ for all $i \neq j \geq 2$. The Gram matrix M of $y_2, \ldots, y_k$ satisfies $M_{ii} = 1$ and $M_{ij} = \frac{1}{2}$ for $i \neq j$, i.e. $M = \frac{1}{2}I + \frac{1}{2}J$ where J is the $(k-1) \times (k-1)$ all-ones matrix. The eigenvalues of M are $\frac{k}{2}$ (once) and $\frac{1}{2}$ ($k-2$ times), all positive, so M is positive definite and has rank $k-1$. Since $y_2, \ldots, y_k \in \mathbb{R}^n$, we need $k-1 \leq n$, i.e. $k \leq n+1$. □

Lemma 0.14 (Bonamy–Kelly–Nelson–Postle, 2022 [7])**.** *There exists an absolute constant $C_0 > 0$ such that the following holds. For all $\omega \geq 3$ and all $\Delta \geq \omega^{C_0}$, every graph G with $\Delta(G) \leq \Delta$ and $\omega(G) \leq \omega$ satisfies*

$$\chi(G) \ \leq \ 72\,\Delta\sqrt{\frac{\ln \omega}{\ln \Delta}}.$$

Proof of Theorem 0.11. By Lemma 0.12, $M(n) = \sup_{\text{finite } X} \chi(G_X)$. Fix a finite unit sphere packing $X \subset \mathbb{R}^n$. By Lemma 0.13, G_X has $\Delta(G_X) \leq \tau(n)$ and $\omega(G_X) \leq n+1$.

We verify the hypothesis of Lemma 0.14 with $\Delta = \tau(n)$ and $\omega = n+2$: since $\tau(n) \geq 2^{0.2075n(1+o(1))}$ by the Kabatiansky–Levenshtein lower bound [4], we have $\ln\tau(n) \geq 0.2075n\ln 2$, while $\ln(n+2)^{C_0} = O(n)$, so $\tau(n) \geq (n+2)^{C_0}$ for all sufficiently large n. Hence

$$\chi(G_X) \;\leq\; 72\,\tau(n)\sqrt{\frac{\ln(n+2)}{\ln\tau(n)}}.$$

Taking the supremum over all finite X gives the first bound.

For the asymptotic form, since $\ln\tau(n) \leq 0.401n\ln 2 + o(n)$ (upper KL bound [4]),

$$\sqrt{\frac{\ln(n+2)}{\ln\tau(n)}} = O\left(\sqrt{\frac{\log n}{n}}\right),$$

giving $M(n) = O(\tau(n)\sqrt{\log n/n}) = 2^{0.401n(1+o(1))}\cdot O(\sqrt{\log n/n})$.

To see this is an improvement over greedy: the ratio of the new bound to $\tau(n)$ is $O(\sqrt{\log n/n}) \to 0$, so the new bound is $o(\tau(n))$ and hence $o(M_{\text{greedy}}(n))$. $\square$

Remark 0.15. The key geometric input is $\omega(G_X) \leq n+1$, which contrasts sharply with $\tau(n) \sim 2^{0.401n}$. The logarithmic ratio $\ln\omega/\ln\Delta \sim (\log n)/(0.401n) \to 0$ is what drives the improvement. The de Bruijn–Erdős step (Lemma 0.12) is essential: the Bonamy–Kelly–Nelson–Postle theorem is a finite-graph result, and the transfer to arbitrary (possibly infinite) packings requires the compactness principle.

Section 7b: Reed's Bound — A Sharper Estimate for Moderate Dimensions

The BKNP bound in Theorem 0.11 is asymptotically the strongest known result, but for moderate dimensions its large absolute constant ($C_0 = 72$) makes it weaker than a classical alternative. We record this alternative, which gives a cleaner closed-form bound superior to BKNP in every dimension n arising in practice.

Theorem 0.16 (Reed's Bound for Sphere Packings)**.** *For all sufficiently large n,*

$$M(n) \;\leq\; \left\lceil \frac{\tau(n)+n+2}{2} \right\rceil.$$

Proof. By Lemma 0.12, it suffices to bound $\chi(G_X)$ for every finite packing tangency graph G_X. By Lemma 0.13, G_X has $\Delta(G_X) \leq \tau(n)$ and $\omega(G_X) \leq n+1$. Reed's theorem [8] states that for every graph G with $\Delta(G) \geq \Delta_0(\omega(G))$ (a threshold depending on the clique number),

$$\chi(G) \;\leq\; \left\lceil \frac{\Delta(G)+\omega(G)+1}{2} \right\rceil.$$

Since $\tau(n) \geq 2^{0.2075n}$ grows exponentially while $\omega(G_X) \leq n+1$ is linear, the condition $\Delta \geq \Delta_0(\omega)$ holds for all sufficiently large n. Substituting $\Delta \leq \tau(n)$

and $\omega \leq n+1$ gives

$$\chi(G_X) \ \leq \ \left\lceil \frac{\tau(n)+n+2}{2} \right\rceil .$$

Taking the supremum over all finite X yields the result. $\square$

Remark 0.17 (Comparison of the Two Upper Bounds)**.** Theorem 0.11 (BKNP) gives $M(n) \leq C\tau(n)\sqrt{\log n/n}$, while Theorem 0.16 gives $M(n) \leq \tau(n)/2+O(n)$. For large n:

$$\text{BKNP} \ \sim \ C\tau(n)\sqrt{\frac{\log n}{n}} \ = \ o(\tau(n)), \qquad \text{Reed} \ \sim \ \frac{\tau(n)}{2}.$$

BKNP is asymptotically stronger ($o(\tau(n))$ vs. $\tau(n)/2$), but the constant $C = 72$ means Reed's bound dominates in every dimension accessible by computation. Concretely, for $n \leq 10^5$ and $\tau(n) \leq 2^{0.401n}$, Reed gives $M(n) \leq \tau(n)/2 + O(n)$, whereas BKNP gives $72\tau(n)\sqrt{\log n/n} > \tau(n)/2$ (since $72^2 \log n/n > 1/4$ for $n \leq 10^5$). The crossover where BKNP beats Reed occurs around $n \approx 4\ln(144^2)\cdot n/\ln n$, i.e. n of order 10^6 or larger.

The combined best-known upper bound is therefore:

$$M(n) \ \leq \ \min\!\left(\left\lceil \tfrac{\tau(n)+n+2}{2} \right\rceil, \ C\tau(n)\sqrt{\tfrac{\log n}{n}} \right).$$

For $n \leq 10^5$ the Reed bound dominates; for $n \to \infty$ the BKNP bound dominates.

Selected values using exact kissing numbers where known and $\tau(n) \leq 2^{0.401n}$ otherwise:

n	$\tau(n)$	Greedy $\tau+1$	Reed $\lceil(\tau+n+2)/2\rceil$	Improvement factor
3	12	13	9	1.44×
4	24	25	15	1.67×
8	240	241	125	1.93×
24	196560	196561	98293	$\approx 2\times$
$n \to \infty$	$\tau(n)$	$\tau(n)+1$	$\tau(n)/2+O(n)$	$\to 2\times$

Reed's bound thus cuts the greedy upper bound by a factor approaching 2, independent of n. The BKNP bound cuts it by a factor $\sqrt{n/\log n} \to \infty$, but only asymptotically.

Remark 0.18 (Barriers to Further Improvement)**.** Both bounds above ultimately rely on $\tau(n)$, the kissing number, as the main input. Any improvement of the form $M(n) \leq \tau(n)^{1-c}$ for a fixed $c > 0$ would require structural properties of contact graphs beyond the degree and clique bounds used here. Candidate approaches include:

- **Lovász theta / SDP**: The Lovász ϑ-function of the contact graph is related to the Cohn–Elkies LP bound for sphere packings, and might yield $M(n) \leq \tau(n)/n^{\varepsilon}$ for some $\varepsilon > 0$.

- **Entropy compression in neighborhoods**: Since each neighborhood $N(v)$ is itself a kissing configuration (a unit sphere packing on S^{n-1}), the local chromatic structure is recursive and may admit a stronger coloring argument.

- **Improving $\tau(n)$**: Any improvement to the Kabatiansky–Levenshtein kissing number bound would immediately improve both $M(n)$ bounds above. No improvement has been found since 1978.

For the *lower* bound, the barrier theorem (Section 6) shows the existing spectral framework is stuck at surplus $O(n^{2/3})$; beating this requires a new geometric construction. Consequently, the gap between the polynomial lower bound $M(n) \geq n+\Omega(n^{2/3})$ and the exponential upper bound $M(n) \leq O(\tau(n))$ remains the central open problem.

Section 8: Further Upper Bounds and Open Problems

We record two additional upper bounds and discuss the principal obstruction to further improvement.

Notation. Write $\tau^+(n)$ for the *one-sided kissing number*: the maximum number of non-overlapping unit spheres whose centres lie in a fixed closed half-space of $\mathbb{R}^n$ and all touch a fixed central unit sphere. Known values: $\tau^+(1) = 1$, $\tau^+(2) = 4$, $\tau^+(3) = 9$ (see [2]). In general, $\tau^+(n) < \tau(n)$ and $\tau^+(n)/\tau(n) \to 1$ as $n \to \infty$; for small n, $\tau^+(n)/\tau(n) = 2/3, 3/4, \ldots$, suggesting $\tau^+(n) \approx \tau(n) \cdot (1 - 1/(n+1))$.

Theorem 0.19 (Brooks Bound and Degeneracy Bound). (i) *For all $n \geq 2$: $M(n) \leq \tau(n)$.*

(ii) *For all $n \geq 1$: $M(n) \leq \tau^+(n) + 1$.*

Proof. Both bounds use Lemma 0.12 (compactness) and Lemma 0.13 (geometric parameters).

(i). By Brooks' theorem [9], a connected graph G satisfies $\chi(G) \leq \Delta(G)$ unless G is a complete graph $K_{\Delta+1}$ or an odd cycle. The tangency graph G_X has $\Delta(G_X) \leq \tau(n)$. For $n \geq 2$, any complete subgraph $K_m \subseteq G_X$ requires m mutually tangent sphere centres, which form an equidistant set in $\mathbb{R}^n$ of size m; since the maximum equidistant set in $\mathbb{R}^n$ has size $n+1$, we have $\omega(G_X) \leq n+1$. For $n \geq 2$, $n+1 < \tau(n)+1$ (since $\tau(2) = 6 > 3$, $\tau(3) = 12 > 4$, etc.), so G_X is not a complete graph $K_{\tau(n)+1}$. Also G_X is not an odd cycle, since for $n \geq 2$ any non-trivial sphere packing has vertices of degree ≥ 1; in any connected component that is not a single edge (and single edges have $\chi = 2 \leq \tau(n)$), the maximum degree exceeds 2 and the graph is not a cycle. Therefore $\chi(G_X) \leq \tau(n)$, giving $M(n) \leq \tau(n)$.

(ii). Fix a unit vector $u \in \mathbb{R}^n$ generic (avoiding any direction parallel to an edge of G_X). Order the vertices of G_X by the linear functional $\ell_u(v) = \langle v, u \rangle$; this defines a total order $v_1 < v_2 < \cdots < v_N$ on the packing centres.

For each v_i, the *back-neighbours* are $B(v_i) := \{v_j : j < i, \{v_i, v_j\} \in E(G_X)\}$, i.e. the neighbours of v_i with a smaller ℓ_u-value. These back-neighbours are sphere centres at distance 1 from v_i with $\langle w - v_i, u\rangle < 0$, lying in the open half of the unit sphere $S^{n-1}(v_i)$ facing $-u$. This is exactly a one-sided kissing configuration around v_i in the half-space $\{w : \langle w - v_i, u\rangle \le 0\}$, so $|B(v_i)| \le \tau^+(n)$.

Greedy colouring in the order $v_1, v_2, \dots$ uses at most $\max_i |B(v_i)| + 1 \le \tau^+(n) + 1$ colours. By Lemma 0.12, $M(n) \le \tau^+(n) + 1$. □

Remark 0.20. Bound (i) (Brooks) improves greedy by exactly 1: $M(n) \le \tau(n)$ vs. $M(n) \le \tau(n)+1$. For $n = 3$: Brooks gives $M(3) \le 12$, Reed gives $M(3) \le 9$, so Reed dominates.

Bound (ii) (degeneracy via half-space kissing) is a new geometric bound. For small n where $\tau^+(n)$ is known:

n	$\tau(n)$	$\tau^+(n)$	Greedy	Bound (ii)	Reed
2	6	4	7	5	5
3	12	9	13	10	9

For $n = 2$: bound (ii) matches Reed. For $n = 3$: Reed is slightly tighter. Asymptotically, since $\tau^+(n) < \tau(n)$ but $\tau^+(n)/\tau(n) \to 1$, bound (ii) is asymptotically dominated by Reed's $\tau(n)/2$.

The interest in bound (ii) is geometric rather than numerical: it expresses the chromatic upper bound purely in terms of the one-sided kissing number, a combinatorial-geometric invariant of $\mathbb{R}^n$ that may be tractable via SDP / spherical code methods independently of the full kissing number.

Summary of all upper bounds. The complete hierarchy for $M(n)$:

$$M(n) \ \le\ \min\Big(\tau^+(n) + 1,\ \tau(n),\ \big\lceil \tfrac{\tau(n)+n+2}{2} \big\rceil,\ 72\,\tau(n)\sqrt{\tfrac{\log n}{n}}\Big).$$

For known small dimensions ($n = 2, 3$): the minimum is achieved by Reed for $n = 3$ and by bound (ii) for $n = 2$; for $n \to \infty$, BKNP dominates.

Open problems.

(1) *Polynomial factor improvement*: Does $M(n) \le \tau(n)/n^{\varepsilon}$ hold for some $\varepsilon > 0$ and all large n? The BKNP bound gives $M(n) \le \tau(n) \cdot O(\sqrt{\log n/n})$ (which is $\tau(n)/n^{1/2-o(1)}$ if log is base 2), so the answer is yes with $\varepsilon = 1/2 - o(1)$ from BKNP. Can one do better, e.g. ε close to 1?

(2) *Contact-graph structure beyond* (Δ, ω): Every contact graph is a *1-distance graph* (edges at distance exactly 1, non-edges at distance > 1 in $\mathbb{R}^n$). This geometric constraint is not captured by Δ and ω alone. In particular, the Gram matrix of the edge vectors is positive semidefinite with a specific rank-n structure. Can this algebraic structure be used in a coloring argument?

(3) *One-sided kissing number*: Determine $\tau^+(n)$ for $n \geq 4$. If $\tau^+(n) \leq (1-c)\tau(n)$ for some absolute constant $c > 0$, then bound (ii) gives $M(n) \leq (1-c)\tau(n) + 1$, a constant-factor improvement independent of n. Current evidence ($\tau^+(2)/\tau(2) = 2/3$, $\tau^+(3)/\tau(3) = 3/4$) is consistent with $\tau^+(n)/\tau(n) \to 1$, in which case bound (ii) is asymptotically equivalent to the greedy bound.

(4) *Lower bound improvement*: Close the gap $n+\Omega(n^{2/3}) \leq M(n) \leq 2^{0.401n(1+o(1))}$. The barrier theorem (Section 6) rules out a spectral improvement for the four classical families; a new geometric construction with chromatic surplus $\omega(n^\alpha)$ for $\alpha > 2/3$ would be needed.

Section 9: Exact Chromatic Number and Theta-Perfectness of the GQ Construction

In this section we sharpen the lower-bound results of Sections 3–4 by determining the chromatic number of G_q exactly. We also compute the Lovász theta function of G_q and its complement, establishing that G_q is *theta-perfect*. These results show the spectral and SDP bounds are simultaneously tight for the GQ family.

9a. Lovász Theta Function for Strongly Regular Graphs

Lemma 0.21 (Lovász theta via eigenvalues)**.** *Let H be a k-regular graph on n vertices with smallest adjacency eigenvalue $\tau < 0$. Then*

$$\vartheta(H) \;=\; \frac{n(-\tau)}{k-\tau}.$$

Moreover, for any graph H, the Lovász sandwich theorem *gives $\omega(H) \leq \vartheta(\overline{H}) \leq \chi(H)$, so $\vartheta(\overline{H})$ is simultaneously a lower bound on $\chi(H)$ and an upper bound on $\omega(H)$.*

Proof. The formula $\vartheta(H) = n(-\tau)/(k-\tau)$ for k-regular H is due to Lovász [10] (see also the matrix formulation in [2]): the optimal solution of the Lovász SDP for $\vartheta(H)$ is achieved by the *Hoffman matrix* $(1/n)(J - (k/\tau)I)$, which is feasible and attains the value $n(-\tau)/(k-\tau)$. The sandwich theorem $\omega(H) \leq \vartheta(\overline{H}) \leq \chi(H)$ follows from the fundamental properties of ϑ [10]: $\alpha(H) \leq \vartheta(H)$ (independence bound), applied to $\overline{H}$ gives $\omega(H) = \alpha(\overline{H}) \leq \vartheta(\overline{H})$. The upper bound $\vartheta(\overline{H}) \leq \chi(H)$ follows from the fractional chromatic duality $\vartheta(\overline{H}) \leq \chi_f(H) \leq \chi(H)$ [10]. $\square$

9b. Theta Values for the GQ Family

Theorem 0.22 (Theta values for G_q and Γ_q)**.** *For every prime power $q \geq 2$, with G_q the complement of the collinearity graph Γ_q of $\mathrm{GQ}(q, q^2)$:*

(a) $\vartheta(G_q) = q + 1 = \alpha(G_q)$.

(b) $\vartheta(\Gamma_q) = \vartheta(\overline{G_q}) = q^3 + 1$.

(c) $\vartheta(G_q) \cdot \vartheta(\Gamma_q) = (q+1)(q^3+1) = v_q$ *(Lovász product equality).*

(d) $\omega(G_q) = \vartheta(\overline{G_q}) = \chi_f(G_q) = q^3 + 1$.

Proof. **(a).** Recall from Section 3 that G_q has $v_q = (q+1)(q^3+1)$ vertices, is $k_q = q^4$-regular, and has smallest adjacency eigenvalue $s_q = -q$. By Lemma 0.21:

$$\vartheta(G_q) \;=\; \frac{v_q \cdot q}{q^4 + q} \;=\; \frac{(q+1)(q^3+1)\cdot q}{q(q^3+1)} \;=\; q+1.$$

Since G_q is vertex-transitive (its automorphism group acts transitively on vertices, as $\mathrm{GQ}(q, q^2)$ is point-transitive), the Lovász theta satisfies $\vartheta(G_q) \geq \alpha(G_q)$. Now $\alpha(G_q) = \omega(\Gamma_q) =$ the clique number of the collinearity graph $=$ the maximum number of mutually collinear points $=$ the line size $q+1$ (since any three mutually collinear points in a GQ lie on a common line, and each line has $q+1$ points). Hence $\alpha(G_q) = q+1 = \vartheta(G_q)$, so the bound is tight.

(b). The collinearity graph Γ_q is $\kappa_q = q(q^2+1)$-regular with smallest eigenvalue $s_0 = -(q^2+1)$ (Section 3). By Lemma 0.21:

$$\vartheta(\Gamma_q) \;=\; \frac{v_q(q^2+1)}{q(q^2+1)+(q^2+1)} \;=\; \frac{v_q(q^2+1)}{(q^2+1)(q+1)} \;=\; \frac{v_q}{q+1} \;=\; q^3+1.$$

(c). $\vartheta(G_q) \cdot \vartheta(\Gamma_q) = (q+1)(q^3+1) = v_q$. For vertex-transitive graphs, Lovász's product theorem gives $\vartheta(H) \cdot \vartheta(\overline{H}) = |V(H)|$ [10], and here $\overline{G_q} = \Gamma_q$.

(d). The clique number of G_q equals the independence number of Γ_q: $\omega(G_q) = \alpha(\Gamma_q)$. By the Hoffman bound applied to Γ_q: $\alpha(\Gamma_q) \leq v_q(-s_0)/(\kappa_q - s_0) = q^3+1$. Ovoids of $\mathrm{GQ}(q, q^2)$ (sets of q^3+1 mutually non-collinear points, one from each line of the GQ) achieve this bound, so $\alpha(\Gamma_q) = q^3+1$ and thus $\omega(G_q) = q^3+1$.

For the fractional chromatic number: G_q is vertex-transitive, so $\chi_f(G_q) = v_q/\alpha(G_q) = (q+1)(q^3+1)/(q+1) = q^3+1$.

By the Lovász sandwich theorem (Lemma 0.21):

$$q^3+1 \;=\; \omega(G_q) \;\leq\; \vartheta(\overline{G_q}) \;=\; \vartheta(\Gamma_q) \;=\; q^3+1.$$

Hence $\omega(G_q) = \vartheta(\overline{G_q}) = q^3+1$; combined with $\chi_f(G_q) = q^3+1$, all four quantities coincide. □

9c. Spread Coloring and Exact Chromatic Number

Definition 0.23. A *spread* of a generalized quadrangle $\mathrm{GQ}(s,t)$ is a set of $st+1$ mutually non-concurrent lines that partition all $(s+1)(st+1)$ points (each point lies on exactly one line of the spread).

For $\mathrm{GQ}(q, q^2)$, a spread consists of $q \cdot q^2 + 1 = q^3+1$ lines of $q+1$ points each, covering all $(q+1)(q^3+1)$ points.

Lemma 0.24 (Existence of a spread in $\mathrm{GQ}(q, q^2)$)**.** *$\mathrm{GQ}(q, q^2)$ has a spread for every prime power $q \geq 2$.*

Proof. Let $\mathrm{GQ}(q^2, q)$ be the dual of $\mathrm{GQ}(q, q^2)$ (in which the roles of points and lines are interchanged). An *ovoid* of $\mathrm{GQ}(q^2, q)$ is a set of $q^3 + 1$ mutually non-collinear points meeting every line in exactly one point. Under the point-line duality of GQs, an ovoid of $\mathrm{GQ}(q^2, q)$ corresponds to a spread of $\mathrm{GQ}(q, q^2)$.

The dual $\mathrm{GQ}(q^2, q)$ is isomorphic to the Hermitian variety $H(3, q^2)$ in $\mathrm{PG}(3, q^2)$ [5]. $H(3, q^2)$ has an ovoid for every prime power q: a classical construction is to take the fixed-point set of a unitary polarity of $\mathrm{PG}(1, q^6)$ restricted to the quadric, giving q^3+1 mutually non-collinear points [5]. Hence $\mathrm{GQ}(q, q^2)$ has a spread. $\square$

Theorem 0.25 (Exact chromatic number)**.** *For every prime power $q \geq 2$,*

$$\chi(G_q) \;=\; q^3 + 1.$$

In particular the Hoffman bound (Section 2) is sharp for G_q.

Proof. **Lower bound:** $\chi(G_q) \geq \chi_f(G_q) = q^3 + 1$ by Theorem 0.22(d).

Upper bound: Fix a spread $\mathcal{S} = \{\ell_1, \ldots, \ell_{q^3+1}\}$ of $\mathrm{GQ}(q, q^2)$ (Lemma 0.24). Each line ℓ_i has $q + 1$ points, and the lines are pairwise disjoint and cover all $v_q = (q + 1)(q^3 + 1)$ points. Define the coloring $c : V(G_q) \to \{1, \ldots, q^3 + 1\}$ by $c(x) = i$ where $x \in \ell_i$.

We verify this is a proper coloring of G_q: two vertices x, y are adjacent in G_q if and only if they are *non-collinear* in $\mathrm{GQ}(q, q^2)$. If $c(x) = c(y) = i$, then $x, y \in \ell_i$, so x and y are collinear (both on line ℓ_i), hence *not* adjacent in G_q. Therefore adjacent vertices receive distinct colors: c is a proper (q^3+1)-coloring.

Combined: $\chi(G_q) = q^3 + 1$. $\square$

Corollary 0.26 (Theta-perfectness of G_q)**.** *For every prime power $q \geq 2$, the graph G_q satisfies*

$$\omega(G_q) \;=\; \vartheta(\overline{G_q}) \;=\; \chi_f(G_q) \;=\; \chi(G_q) \;=\; q^3 + 1.$$

All four graph parameters coincide; equivalently, G_q attains the Hoffman bound with equality at both the fractional and integral chromatic levels.

Remark 0.27. The equality $\omega(G_q) = \chi(G_q)$ does not mean G_q is a perfect graph (the perfect graph theorem concerns all induced subgraphs, not just the whole graph). What is remarkable is that the *spectral* lower bound (Hoffman), the *SDP* lower bound ($\vartheta(\overline{G_q})$), the *fractional* chromatic number, and the integer chromatic number all give the same value $q^3 + 1$. This mutual tightness is unusual: typically Hoffman $\leq \vartheta(\overline{G}) \leq \chi_f \leq \chi$ is strict at one or more steps.

9d. Sharpened Packing Lower Bound

Combining Theorem 0.25 with the Spectral Realization Theorem (Section 1):

Corollary 0.28 (Sharp lower bound at d_q)**.** *For every prime power $q \geq 2$, the unit sphere packing in $\mathbb{R}^{d_q}$ realizing G_q has chromatic number exactly q^3+1. In particular $M(d_q) \geq q^3+1$, and a packing achieving this lower bound is explicitly constructible via the spread coloring of $\mathrm{GQ}(q, q^2)$.*

9e. Implications for the Linear Upper Bound Conjecture

The exact results above provide quantitative evidence for a much stronger upper bound on $M(n)$ than the current exponential bound.

Conjecture 0.29 (Linear upper bound)**.** There exists an absolute constant C such that $M(n) \leq Cn$ for all $n \geq 1$.

The GQ construction shows that $C \geq 1$ is necessary ($M(d_q)/d_q \to 1$), and more precisely that the best possible linear constant satisfies $C = 1$ with a sub-linear surplus:

$$M(n) \;=\; n + \Theta(n^{2/3}) \quad \text{(strong form of the conjecture)},$$

consistent with the lower bound $M(d_q) \geq d_q + (3/4)^{1/3} d_q^{2/3}$ from Section 4.

The following observations provide structural evidence:

Remark 0.30 (Evidence from the Lovász theta function)**.** Corollary 0.26 shows $\vartheta(G_q) = q + 1 = O(d_q^{1/3})$: the Lovász theta of the sphere packing contact graph G_q in $\mathbb{R}^{d_q}$ is only $O(n^{1/3})$ rather than $O(n)$. This means:

1. The independence number satisfies $\alpha(G_q) = \vartheta(G_q) = q + 1$ (the SDP bound for α is tight for G_q).
2. The fractional chromatic number satisfies $\chi_f(G_q) = v_q/\alpha(G_q) = q^3 + 1 = O(d_q)$, consistent with $M(n) = O(n)$.
3. More broadly, the key invariant bounding χ_f is the ratio v/α, which equals $q^3 + 1 = O(d_q)$ for the extremal GQ family. If every sphere packing contact graph G in $\mathbb{R}^n$ satisfies $v(G)/\alpha(G) = O(n)$, then $\chi_f(G) = O(n)$ and Conjecture 0.29 follows.

Thus the conjecture is equivalent to: *For every sphere packing contact graph G in $\mathbb{R}^n$, $\alpha(G) \geq |V(G)|/(Cn)$ for some absolute constant C.*

Remark 0.31 (Polynomial bounds via clique structure)**.** A weaker but still significant consequence of the exact results: combining $\omega(G_q) = q^3 + 1 = O(d_q)$ and the BKNP bound (Theorem 0.11) with $\omega = O(n)$ and $\Delta = \tau(n)$:

$$\chi(G_q) \;\leq\; 72\,\tau(d_q)\sqrt{\frac{\ln(q^3+2)}{\ln \tau(d_q)}} \;=\; O\!\left(\tau(d_q)\sqrt{\frac{\log d_q}{d_q}}\right),$$

which matches the general bound. The gap between this and the exact value $q^3 + 1 \approx d_q$ is a factor of approximately $\tau(d_q)/d_q = 2^{0.401\,d_q}/d_q$, an exponential overcount. Closing this gap requires exploiting the geometric sphere-packing constraint beyond the degree and clique bounds.

Remark 0.32 (All known examples satisfy $\chi(G) \leq 2n$)**.** For the GQ construction: $\chi(G_q) = q^3 + 1 \leq 2(q^3 - q^2 + q) = 2d_q$ for all $q \geq 2$ (since $q^3 + 1 \leq 2q^3 - 2q^2 + 2q$ iff $(q-1)(q^2 - q + 1) \geq 0$, which holds for all $q \geq 1$).

For the A_n lattice packing: $\chi = n + 1 \le 2n$ for $n \ge 1$. The simplex packing: $\chi = n + 1 \le 2n$. All other known constructions satisfy $\chi \le n + 1$. We know of no sphere packing contact graph G in $\mathbb{R}^n$ with $\chi(G) > 2n$.

Corollary 0.33 (Main Result)**.** *The supremum $M(n)$ of chromatic numbers of unit sphere packings in $\mathbb{R}^n$ satisfies:*

(i) ***Lower bound:***

$$\limsup_{n\to\infty} \frac{M(n) - n}{n^{2/3}} \ \ge\ 1.$$

In particular, $(M(n) - n)/n^{2/3}$ is at least $(1 - 1/q + 1/q^2)^{1/3} \to 1$ along $n = d_q = q^3 - q^2 + q$ as $q \to \infty$ through prime powers.

(ii) ***Upper bound:***

$$M(n) \ =\ O\left(\tau(n) \cdot \sqrt{\frac{\log n}{n}}\right) \ =\ 2^{0.401n(1+o(1))} \cdot O\left(\sqrt{\frac{\log n}{n}}\right).$$

(iii) ***Barrier:*** *By the Barrier Theorem* (Theorem 0.9)*, none of the four named classical spectral families from finite geometry (Kneser, triangular, symplectic polar, hexagon complement) achieves a positive surplus $H(G) - d > 0$ within the Hoffman framework.*

Proof. **Lower bound chain.** By the Spectral Realization Theorem (Theorem 0.1) and the explicit spectral computation in Theorem 0.3, the complement G_q of the collinearity graph of $\mathrm{GQ}(q, q^2)$ embeds as the tangency graph of a unit sphere packing in $\mathbb{R}^{d_q}$, so

$$M(d_q) \ \ge\ \chi(G_q).$$

By the Hoffman Bound (Theorem 0.2) and the eigenvalues $k_q = q^4$, $s_q = -q$ of G_q,

$$\chi(G_q) \ \ge\ 1 - \frac{k_q}{s_q} = q^3 + 1.$$

Hence $M(d_q) \ge q^3 + 1 = d_q + (q^2 - q + 1)$ (Theorem 0.4).

Limsup. By Theorem 0.5(2), taking $q \to \infty$ through prime powers,

$$\frac{M(d_q) - d_q}{d_q^{2/3}} \ \ge\ \left(1 - \frac{1}{q} + \frac{1}{q^2}\right)^{1/3} \ \to\ 1,$$

which gives $\limsup_{n\to\infty}(M(n) - n)/n^{2/3} \ge 1$.

Upper bound. This is Theorem 0.11.

Barrier. This is Theorem 0.9. $\square$

References

[1] H. Chen, *Ball packings with high chromatic numbers from strongly regular graphs*, Discrete Mathematics **340** (2017), no. 7. arXiv:1502.02070.

[2] A. E. Brouwer and H. Van Maldeghem, *Strongly Regular Graphs*, Cambridge University Press, 2022.

[3] A. J. Hoffman, *On eigenvalues and colorings of graphs*, in Graph Theory and its Applications (B. Harris, ed.), Academic Press, 1970, pp. 79–91.

[4] G. A. Kabatiansky and V. I. Levenshtein, *Bounds for packings on the sphere and in space*, Problemy Peredachi Informatsii **14** (1978), 3–25.

[5] S. E. Payne and J. A. Thas, *Finite Generalized Quadrangles*, Research Notes in Mathematics **110**, Pitman, Boston, 1984. Second edition: EMS Series of Lectures in Mathematics, European Mathematical Society, Zürich, 2009.

[6] N. G. de Bruijn and P. Erdős, *A colour problem for infinite graphs and a problem in the theory of relations*, Indagationes Mathematicae **13** (1951), 369–373.

[7] M. Bonamy, T. Kelly, P. Nelson, and L. Postle, *Bounding χ by a fraction of Δ for graphs without large cliques*, Journal of the European Mathematical Society **26** (2024), no. 5, 1451–1499. arXiv:1803.01051.

[8] B. Reed, *ω, Δ, and χ*, Journal of Combinatorial Theory, Series B **74** (1998), 241–260.

[9] R. L. Brooks, *On colouring the nodes of a network*, Proceedings of the Cambridge Philosophical Society **37** (1941), 194–197.

[10] L. Lovász, *On the Shannon capacity of a graph*, IEEE Transactions on Information Theory **25** (1979), 1–7.